

CongestiQ

AI-Powered Port Congestion Prediction & Operations Optimizer

IBM Bob AI Hackathon 2026



Predict congestion. Optimize operations. Keep cargo moving.

Problem



- Port operators manage hundreds of vessels with limited berth and crane capacity.
- Congestion is often identified only after vessel queues have already formed.
- Manual allocation of berths, cranes, and yard space makes proactive planning difficult.
- Delays can cause berth conflicts, inefficient resource utilization, and supply-chain disruption.
- Alternative routing and operational decisions may come too late to reduce congestion effectively
- **Ports need a forward-looking system that predicts congestion and turns those predictions into actionable operational decisions.**





Current Challenges

- **Unpredictable Port Congestion** — Congestion can build rapidly and create long vessel queues.
- **Manual Resource Allocation** — Berths, cranes, and yard space must be coordinated across many vessels.
- **Late Detection of Congestion** — Operators may identify congestion only after queues have already formed.
- **Vessel Delays & Berth Conflicts** — Poor coordination can increase waiting time and resource conflicts.
- **Difficult Alternative Routing** — Late congestion detection makes alternative routing decisions harder.

Solution



- Predict upcoming port congestion using vessel schedules and port capacity.
 - Identify high-risk vessels and congestion hotspots.
 - Optimize berth and crane assignments based on operational constraints.
 - Recommend alternative operational actions for high-risk vessels.
 - Plan a prioritized 72-hour operations plan for port supervisors.
 - Provide all insights through a single React dashboard.
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- **CongestiQ transforms congestion prediction into proactive operational decision-making.**



How it works

- CongestiQ transforms vessel and port resource data into congestion predictions, operational optimization, and actionable 72-hour planning through a single integrated dashboard.

 Vessel Schedules

 Port Resources



 Congestion Prediction



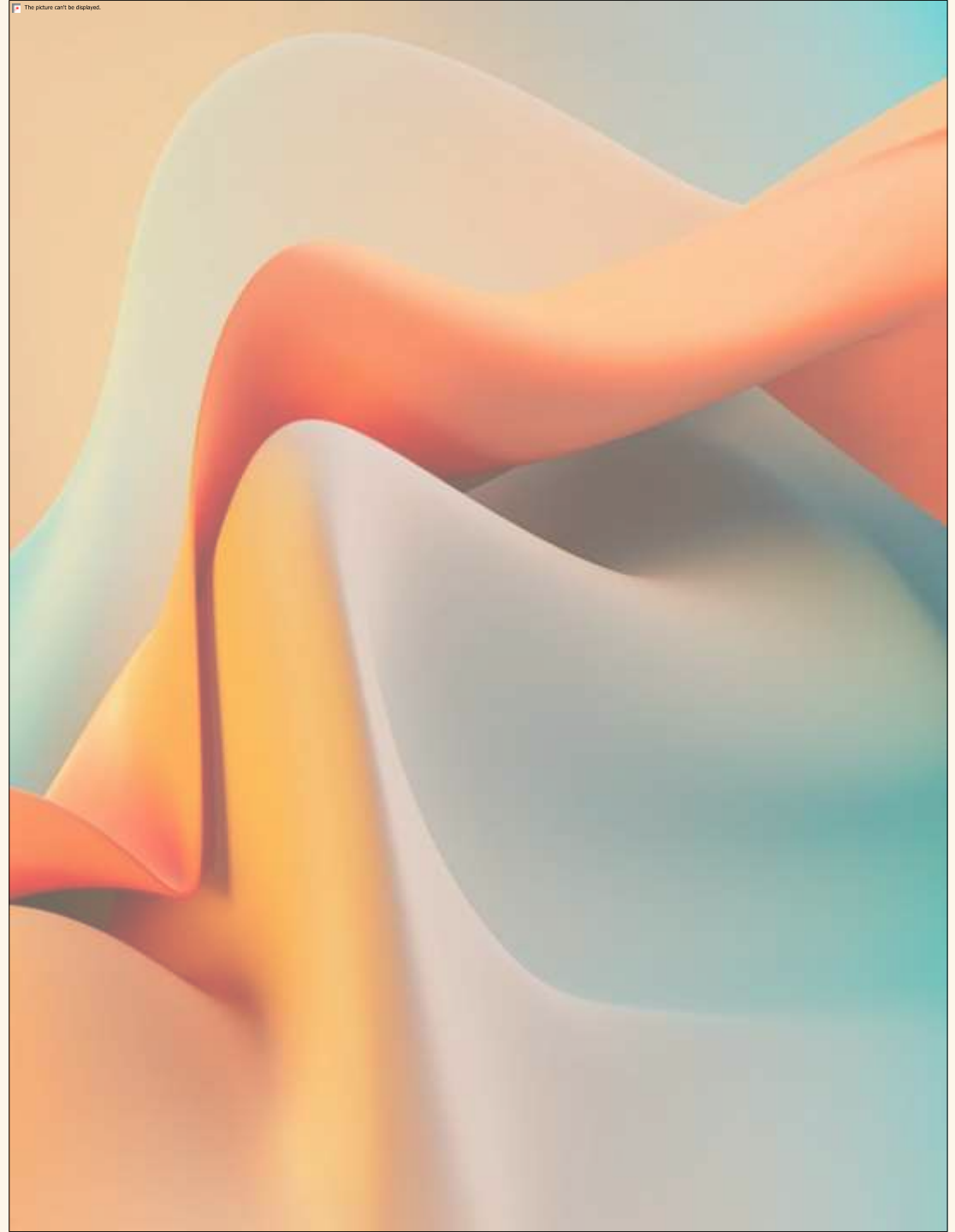
 Risk + Hotspots



 Optimization



 72-Hour Plan

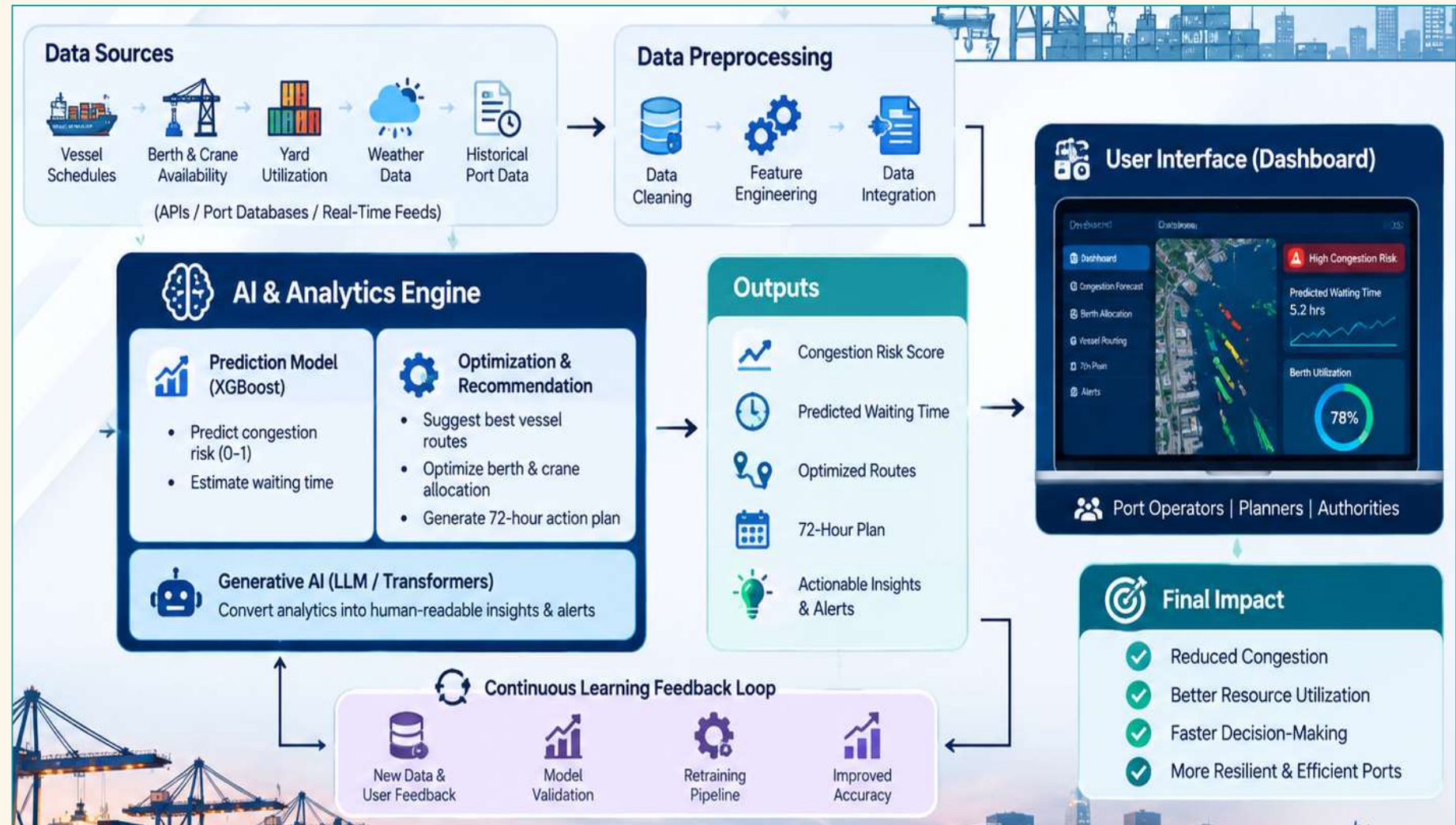


Technical Approach

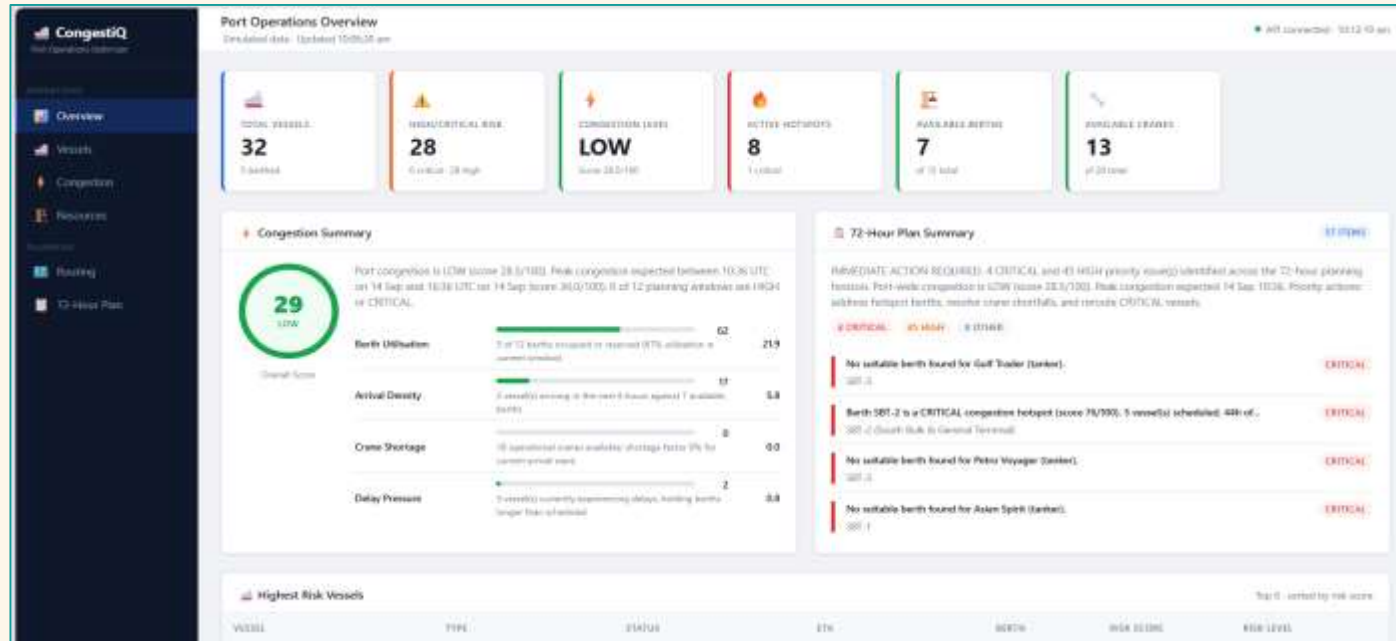


Tech Stack Used

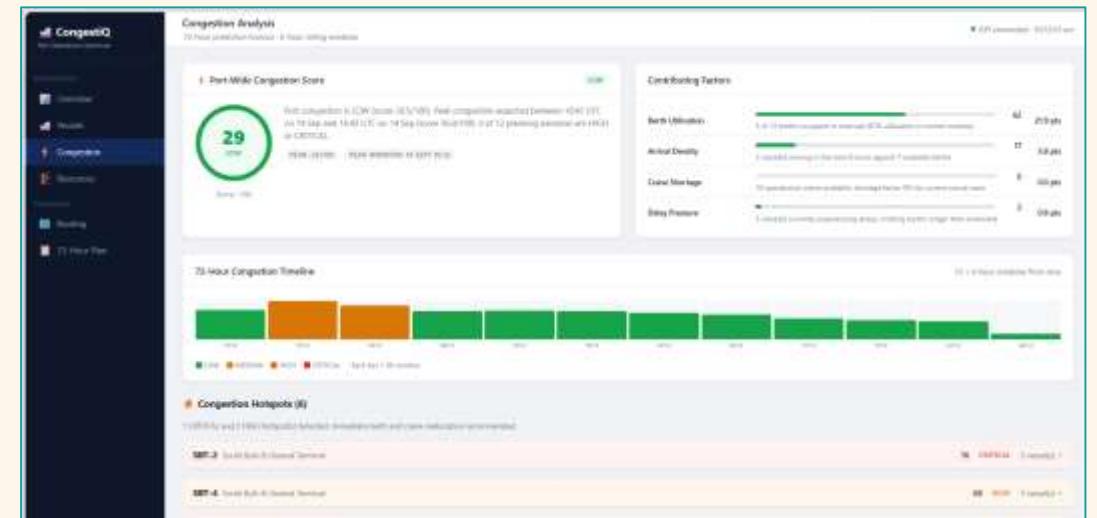
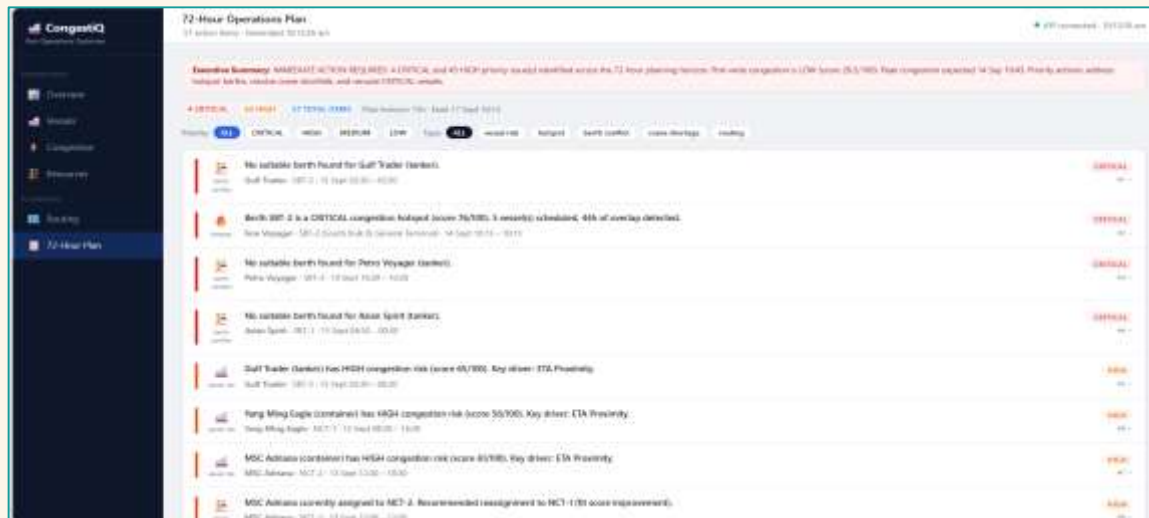
- React + Vite
- Python
- FastAPI
- SQLite / PostgreSQL
- REST APIs
- JavaScript
- IBM Bob



Live Application



Live Dashboard Congestion Prediction 72-Hour Operations Plan





IBM Bob Integration

How IBM Bob was used in CongestiQ

-  **AI-assisted development** of the CongestiQ application
-  Generated and refined **backend APIs**
-  Assisted with **congestion prediction and risk-analysis** logic
-  Assisted with **optimization and routing components**
-  Supported development of the **React dashboard**
-  Used for **debugging, testing, and iterative improvements**

Our Workflow with IBM Bob



- “With the help of IBM Bob, we turned our ideas into a working solution — faster, smarter, and more efficiently.”

Impact & Benefits



• Expected Impact

- Earlier identification of congestion risks
- Better berth and crane utilization
- Faster operational decision-making
- Reduced vessel-delay risk
- Clearer prioritization of near-term actions
- Improved visibility into upcoming port congestion

• Future Scope

- Integrate live AIS and vessel schedule data
- Connect to real-time berth and crane availability
- Use historical port data for improved prediction
- Add real-time operational alerts
- Add authentication and role-based access
- Extend to multi-port optimization





CongestiQ

Predict | Optimize | Plan | Keep Ports Moving

Powered by
IBM bob
Build Smarter with AI

IBM Bob
AI Hackathon 2026

Thank You

FOR BEING PART OF OUR JOURNEY

Together with IBM Bob,
we turn data into smoother, smarter and
more resilient ports.

CLEANER PORTS
GREENER TOMORROW



PREDICT
CONGESTION RISKS



OPTIMIZE
RESOURCES



ROUTE
VESSELS SMARTLY



PLAN
72-HOUR OPERATIONS

DATA-DRIVEN PORTS
FOR A SMOOTHER TOMORROW

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SMARTER PORTS
STRONGER ECONOMIES
BRIGHTER TOMORROWS